AN OBSTRUCTION TO PRIME-LOCAL FIRST-DIFFERENCE DECOMPOSITIONS OF WEIL'S QUADRATIC FORM

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Abstract. Weil's criterion makes positivity of an explicit quadratic form equivalent to the Riemann hypothesis, which motivates attempts to assemble that form from manifestly positive prime-local pieces. We isolate the prime-power term and test a natural such architecture. For a prime power n, put $w_n = \Lambda(n)/\sqrt{n}$, $a_n = \log n$, and let $T_a\varphi(x) = \varphi(x-a)$. We prove that the scalar-completed local block

$$b_n \|\varphi\|_2^2 - 2w_n \operatorname{Re}\langle\varphi, T_{a_n}\varphi\rangle$$

is nonnegative on $C_c^\infty(\mathbb{R})$ exactly when $b_n \geq 2w_n$. The required scalar budget therefore diverges, since $\sum_{n\geq 2}\Lambda(n)/\sqrt{n} = \infty$. For every fixed nonzero compactly supported test function, the ordinary sum of such positive blocks diverges to $+\infty$. Consequently, the actual prime-power term cannot be assembled as an ordinary convergent sum of individually positive first-difference blocks with a fixed summable scalar completion. Finite truncations and summable synthetic weights remain valid. The obstruction comes from infinite-weight assembly; each local block still has an exact algebraic threshold.

1. The obstruction

Weil's positivity criterion recasts the Riemann Hypothesis as positivity of a quadratic functional derived from the explicit formula [9, 1, 4, 7]. Its prime-power contribution has a particularly simple translation structure. That structure invites a local sum-of-squares construction: assign one positive first-difference block to each prime power, then assemble the blocks.

This paper determines exactly when that construction works. Positivity of the block attached to a prime power of natural weight w_n forces scalar compensation $b_n \geq 2w_n$. For the actual Weil weights, the resulting total compensation is infinite.

Let $C_c^\infty(\mathbb{R})$ carry the inner product

$$\langle\varphi,\psi\rangle = \int_{\mathbb{R}} \varphi(x)\overline{\psi(x)}\,dx, \qquad T_a\varphi(x) = \varphi(x-a).$$

For $\varphi \in C_c^\infty(\mathbb{R})$ define

$$\widetilde{\varphi}(x) = \overline{\varphi(-x)}, \qquad F_\varphi = \varphi * \widetilde{\varphi}.$$

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Then

$$F_\varphi(a) = \langle \varphi, T_a\varphi \rangle\,, \qquad F_\varphi(-a) = \overline{F_\varphi(a)}. \tag{1}$$

Write $\mathcal{P}^* = \{p^k : p \text{ prime},\ k \geq 1\}$ and, for $n \in \mathcal{P}^*$, set

$$w_n = \frac{\Lambda(n)}{\sqrt{n}} > 0, \qquad a_n = \log n > 0.$$

Given a real scalar b_n, consider

$$Q_n^{(b)}(\varphi) = w_n \left\| \varphi - T_{a_n}\varphi \right\|_2^2 + (b_n - 2w_n) \left\| \varphi \right\|_2^2 \tag{2}$$

$$= b_n \left\| \varphi \right\|_2^2 - 2w_n \operatorname{Re} F_\varphi(a_n). \tag{3}$$

Theorem 1.1 (Prime-local first-difference obstruction)**.** *Let $(b_n)_{n\in\mathcal{P}^*}$ be real numbers.*

1. *The block $Q_n^{(b)}$ is nonnegative on $C_c^\infty(\mathbb{R})$ if and only if $b_n \geq 2w_n$.*
2. *If every block is nonnegative, then*
$$\sum_{n\in\mathcal{P}^*} b_n = +\infty.$$
3. *If every block is nonnegative, then for every nonzero $\varphi \in C_c^\infty(\mathbb{R})$,*
$$\sum_{n\in\mathcal{P}^*} Q_n^{(b)}(\varphi) = +\infty.$$

Consequently, no ordinary convergent sum of individually nonnegative blocks of the form (2) can represent the prime-power term with a fixed summable scalar completion.

The three assertions separate the mechanism cleanly. The first is an exact local positivity threshold. The second inserts the arithmetic weights. The third uses compact support to eliminate the correlation tail. The divergent scalar tail remains and prevents positive assembly.

2. The exact local threshold

The threshold follows from the elementary approximate translation invariance of broad normalized plateaux. This local device is standard; the arithmetic content begins when its sharp scalar cost is imposed simultaneously at every actual prime power.

Proposition 2.1. *Fix $a \neq 0$, $w > 0$, and $b \in \mathbb{R}$. The quadratic form*

$$Q_{a,w,b}(\varphi) = w \left\| \varphi - T_a\varphi \right\|_2^2 + (b - 2w) \left\| \varphi \right\|_2^2$$

is nonnegative for every $\varphi \in C_c^\infty(\mathbb{R})$ if and only if $b \geq 2w$.

Proof. If $b \geq 2w$, both terms in the displayed expression are nonnegative.

For the converse, choose a nonzero $\eta \in C_c^\infty(\mathbb{R})$ and set

$$\eta_L(x) = L^{-1/2}\eta(x/L), \qquad L > 0.$$

The normalization gives $\|\eta_L\|_2 = \|\eta\|_2$. The fundamental theorem of calculus and a change of variables give the quantitative estimate

$$\left\| \eta_L - T_a\eta_L \right\|_2 \leq \frac{|a|}{L} \left\| \eta' \right\|_2. \tag{4}$$

Therefore

$$Q_{a,w,b}(\eta_L) \longrightarrow (b-2w)\,\|\eta\|_2^2 \qquad (L\to\infty).$$

Nonnegativity for every L forces $b-2w\geq 0$. □

For each $n\in\mathcal{P}^*$, apply Proposition 2.1 with $a=a_n$, $w=w_n$, and $b=b_n$. This proves Theorem 1.1(1). The proof uses the test space itself: every η_L is smooth and compactly supported, although its support expands with L. No completion of the test space and no spectral approximation are needed.

3. The arithmetic budget diverges

The local threshold becomes a global obstruction because the actual prime-power weights are not summable.

Proposition 3.1. *One has*

$$\sum_{n\in\mathcal{P}^*}\frac{\Lambda(n)}{\sqrt{n}} = +\infty.$$

Proof. Let

$$W_N = \sum_{1\leq n\leq N}\frac{\Lambda(n)}{\sqrt{n}}, \qquad \psi(N)=\sum_{1\leq n\leq N}\Lambda(n).$$

Since $\Lambda(n)\geq 0$ and $\sqrt{n}\leq\sqrt{N}$ on this range,

$$W_N \geq \frac{\psi(N)}{\sqrt{N}}. \tag{5}$$

We use the elementary Chebyshev bound

$$\psi(N)\geq N\log 2-\log(N+1). \tag{6}$$

For completeness, let $L_N=\operatorname{lcm}(1,\dots,N)$. Prime-adic valuation gives $\binom{N}{k}\mid L_N$ for $0\leq k\leq N$; hence

$$2^N=\sum_{k=0}^{N}\binom{N}{k}\leq (N+1)L_N.$$

Taking logarithms and using $\psi(N)=\log L_N$ proves (6). Equations (5) and (6) now give

$$W_N\geq\sqrt{N}\log 2-\frac{\log(N+1)}{\sqrt{N}}.$$

The right-hand side tends to $+\infty$. Thus $W_N\to+\infty$. Finally, $\Lambda(n)$ vanishes away from prime powers, so these are exactly the increasing-cutoff partial sums in the proposition. □

If every $Q_n^{(b)}$ is nonnegative, Proposition 2.1 yields $b_n\geq 2w_n$. Proposition 3.1 therefore proves Theorem 1.1(2).

4. Failure of ordinary positive assembly

Compact support makes the prime-power correlation sum finite, but it makes the positive-block obstruction more transparent.

Proof of Theorem 1.1(3). Fix nonzero $\varphi \in C_c^\infty(\mathbb{R})$. The autocorrelation $F_\varphi = \varphi * \widetilde{\varphi}$ is compactly supported. Hence there is $A > 0$ such that

$$F_\varphi(a_n) = 0 \qquad \text{whenever } a_n > A.$$

For all sufficiently large $n \in \mathcal{P}^*$, equation (3) becomes

$$Q_n^{(b)}(\varphi) = b_n \, \|\varphi\|_2^2 \,.$$

Every term is nonnegative, and the tail sum diverges by part (2). Thus the ordinary series diverges to $+\infty$. ☐

In the Fourier convention

$$h(r) = \int_{\mathbb{R}} F_\varphi(u) e^{iru} \, du, \qquad F_\varphi(u) = \frac{1}{2\pi} \int_{\mathbb{R}} h(r) e^{-iru} \, dr,$$

one has $h(r) \geq 0$ on $\mathbb{R}$. The prime-power part of Weil's explicit formula in this normalization is

$$P(\varphi) = -\sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \big(F_\varphi(\log n) + F_\varphi(-\log n)\big) \tag{7}$$

$$= -2 \sum_{n \in \mathcal{P}^*} w_n \operatorname{Re} F_\varphi(a_n). \tag{8}$$

This is the sign and factor-of-two convention used in [4]; Appendix A records the audit.

Suppose an ordinary blockwise-positive construction tried to realize

$$P(\varphi) + B \, \|\varphi\|_2^2 = \sum_{n \in \mathcal{P}^*} Q_n^{(b)}(\varphi), \qquad B = \sum_{n \in \mathcal{P}^*} b_n < \infty. \tag{9}$$

Equations (3) and (8) give the formal identity, but Theorem 1.1 rules out its positive ordinary assembly: blockwise positivity forces $B = +\infty$, and the right-hand side diverges on every nonzero test function.

Corollary 4.1. *The actual prime-power contribution to Weil's quadratic form admits no decomposition of the form (9) with all of the following properties:*

1. *one uncoupled block is assigned to each prime power;*
2. *each block is a scalar-completed first translation difference as in (2);*
3. *each block is nonnegative on all of $C_c^\infty(\mathbb{R})$; and*
4. *the scalar completion is an ordinary summable series independent of the test function.*

Corollary 4.1 resolves this architecture. Cross-prime couplings, nonlocal quadratic forms, operator-valued completions, test-function-dependent counterterms, and independently justified renormalized sums do not satisfy its hypotheses and require separate analysis. The theorem neither proves nor disproves the Riemann Hypothesis.

5. Controls that isolate the mechanism

Two controls show exactly where the obstruction enters.

Proposition 5.1 (Finite-prefix control)**.** *For every finite set $E \subset \mathcal{P}^*$, the choice $b_n = 2w_n$ gives*

$$\sum_{n\in E} Q_n^{(b)}(\varphi) = \sum_{n\in E} w_n \left\|\varphi - T_{a_n}\varphi\right\|_2^2 \geq 0.$$

Thus every finite prefix has a valid positive first-difference decomposition.

Proposition 5.2 (Summable-weight control)**.** *Let $(c_j)_{j\geq 1}$ be nonnegative with $\sum_j c_j < \infty$, and let $(\alpha_j)_{j\geq 1} \subset \mathbb{R}$. Then*

$$\sum_{j\geq 1} c_j \left\|\varphi - T_{\alpha_j}\varphi\right\|_2^2$$

converges for every $\varphi \in L^2(\mathbb{R})$ and defines a nonnegative quadratic form.

Proof. Translation is unitary, so

$$c_j \left\|\varphi - T_{\alpha_j}\varphi\right\|_2^2 \leq 4c_j \left\|\varphi\right\|_2^2.$$

The asserted convergence follows by comparison with $4\left\|\varphi\right\|_2^2 \sum_j c_j$. □

The finite-prefix control verifies the local algebra against the actual weights. The summable-weight control verifies the infinite assembly against a convergent model. Together they locate the failure in the conjunction of actual prime-power weights, per-block positivity, and ordinary uncoupled summation.

6. Relation to existing formulations

Weil's original criterion established the global positivity framework [9]. Burnol's place-by-place expositions emphasize how local terms enter the explicit formula [2, 3], while Conrey gives a convenient normalization of the zeta explicit formula and its positivity criterion [4]; Lagarias relates Li coefficients to values of Weil's quadratic functional on explicit test functions [7]. Yoshida studied Hermitian forms attached to zeta functions [10]. Bombieri analyzed Weil's quadratic functional, including compact-support restrictions, finite truncations, and associated spectral questions [1]. Recent screw-function formulations organize several positivity approaches through nonlocal operator structures [8]. Finite Galerkin dictionaries and numerical realizations now study truncated or nonlocal Weil operators directly [5, 6].

The present result addresses a different, sharply specified question: whether the prime-power term itself can be assembled from uncoupled, individually positive first-translation-difference blocks using a fixed summable scalar budget. Theorem 1.1 answers that question exactly. Its local threshold is an elementary approximate-invariance calculation; the theorem's content is the sharp application of that threshold to the actual prime-power weights and the resulting infinite-assembly exclusion. The cited global, finite-dimensional, and nonlocal formulations remain outside the excluded class.

Appendix A. Normalization audit

This appendix makes the sign, conjugation, and factor of two independently checkable. Starting from the definitions,

$$F_\varphi(a) = \int_{\mathbb{R}} \varphi(y)\widetilde{\varphi}(a-y)\,dy$$
$$= \int_{\mathbb{R}} \varphi(y)\overline{\varphi(y-a)}\,dy = \langle \varphi, T_a\varphi\rangle\,.$$

Changing variables and conjugating gives $F_\varphi(-a) = \overline{F_\varphi(a)}$. Consequently,

$$-w(F_\varphi(a) + F_\varphi(-a)) = -2w\,\mathrm{Re}\,\langle \varphi, T_a\varphi\rangle\,.$$

Unitarity of T_a gives

$$\|\varphi - T_a\varphi\|_2^2 = \|\varphi\|_2^2 + \|T_a\varphi\|_2^2 - 2\,\mathrm{Re}\,\langle \varphi, T_a\varphi\rangle$$
$$= 2\,\|\varphi\|_2^2 - 2\,\mathrm{Re}\,F_\varphi(a).$$

Multiplying by w and adding $(b-2w)\,\|\varphi\|_2^2$ yields (3). These identities fix every sign and numerical factor used in the main theorem.

Appendix B. Verification remarks

The proof is symbolic and requires no numerical experiment. Its two key calculations are the expansion of the translation-difference norm in (3) and the normalized-dilation estimate (4), which together give the sharp threshold $b \geq 2w$. Proposition 3.1 then follows from the lcm form of Chebyshev's lower bound (6), while compact support of F_φ identifies the tail $Q_n^{(b)}(\varphi) = b_n\,\|\varphi\|_2^2$. The finite-prefix and summable-weight controls in Propositions 5.1 and 5.2 record the adjacent regimes in which the same algebra does assemble correctly.

Data and code availability. The theorem and both controls are proved completely in the text and use no external data or numerical certificate. An accompanying Lean 4 development checks the arithmetic divergence with exact prime-power indexing, the abstract local threshold and scalar obstruction, and both controls. Its stated scope does not include the concrete C_c^∞ dilation argument or the explicit-formula normalization, which remain proved in the text.

Assistance disclosure

AI systems were used extensively in mathematical derivation, Lean proof development, literature discovery, organization, and typesetting. The author remains responsible for every mathematical statement, proof, citation, and submission decision.

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